Math 110 Fall 2026 Running Review Document

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1 Disclaimer

This review document will be based on the lectures of Alan Hammond in the Fall of 2026 (our course). These lectures are themselves based on the lectures of Edward Frenkel, and Sheldon Axler's *Linear Algebra Done Right*. These are supplementary and are not a replacement for the lecture notes, or for attending lecture. They are meant as a short recap with some guiding questions based on my personal understanding of the content, and are not comprehensive.

2 Super-Short Test Review

Fields: What are the field axioms, and what are some common fields? You should also be able to prove some basic properties using the field axioms, like that if $ab = 0$, either $a = 0$ or $b = 0$. Make sure to know $\mathbb{F}_2$.

Vector Spaces: What are the vector space axioms and how do they differ from the field axioms? You should focus on the operations and their properties. Again, some basic properties from the axioms include things like $0v = \vec{0}$ for every v in the vector space. Why does the field that the vector space is over matter? You should be able to show that common things are vector spaces, like $\mathbb{F}^n$, the space of sequences over $\mathbb{F}$, the polynomials, and the matrices.

Subspaces: What is the definition of a subspace, and what tool do we use to make it easier to show something is a subspace? You should know some examples of subspaces and be able to determine whether a given space is or is not a subspace.

Subspace Sums: Why do we take sums instead of unions? What is special about the subspace sum? What are direct sums and how do they sharpen this definition? You should be able to compute sums, look at examples of why this subspace sum is not an invertible operation/doesn't behave like vector sums, and given $U \subset \mathbb{R}^n$ you should be able to find V so $U \oplus V = \mathbb{R}^n$.

Linear Independence and Span: You should be able to define linear independence and use that definition to show that a set of vector is linearly independent. You should be able to define a span, find spanning sets for some vector spaces, and both prove and use the deletion lemma. Combining the two ideas gives a basis, and you should be able to work with bases, do computations with them, and use the building up and cutting down lemmas to manipulate bases.

Dimension: You should know that the dimension is the length of a basis, the formula for the dimension of a sum of subspaces. You should be able to compute the dimension in some cases and reason about subspace inclusion or equality using dimension.

Linear Maps: You should know the definition of a linear map and some basic examples of linear maps. You should recognize and be able to define the null space and image of a linear map, relate these subspaces to the injectivity and surjectivity of the map, and be able to find them for many examples. Lastly, you should know the rank-nullity theorem and use it to reason about whether linear maps are injective, surjective, or both. The rank-nullity theorem will also be used to prove many dimension equalities and inequalities in future sections.

Matrices: You should know what a matrix is, even over weird fields, and be able to connect linear maps to their matrices under a given basis/set of coordinates. You should also be able to move between bases and coordinates, multiply matrices, and be able to connect the vector space $L(V, V)$ to the space of matrices by knowing how operations connect in each space (including composition).

Isomorphisms: You should know the definition of an isomorphism and how to prove that a map is an isomorphism. This includes recognizing inverses, their uniqueness, and computing some inverses. In general, what finite-dimensional spaces are isomorphic? You should also relate injectivity and surjectivity to left and right inverses.

Product Spaces, Quotients: How do we define products? How do products relate to direct sums? You should be able to identify when spaces look like products and use products to build example vector spaces for other problems. Quotient spaces are often tricky for students at first, so be careful about understanding the definition of translates (or cosets) and how translates form a vector spaces. You should be able to do basic calculations with translates. Commit to memory the connection between the quotient space and the image of a linear map, especially how we define the quotient map and connect the dimension of V to the dimension of its quotient space V/U.

Dual Spaces: You should be able to define a linear functional, as well as prove $L(V, \mathbb{F}) \simeq V$ for finite-dimensional V. Given a basis, you should be able to construct the dual basis and use it for calculations and proofs (such as connecting it to the coordinates of a vector). Given a map, you should be able to find its dual map both as an abstract definition and for some maps- in particular, you should be able to connect the

dual map and transpose where applicable. You should also be able to connect the null space and range of a linear map to those of its dual using the annihilator, and know apply dimensions in both cases.

Invariant Subspaces: You should be able to define and calculate eigenvalues, and their associated eigenvectors/eigenspaces. You should connect the eigenvectors to invariant subspaces.

The Minimal Polynomial: You should be able to define the minimal polynomial and justify that it always exists. You should be able to compute the polynomial itself, as well as do calculations with the minimal polynomial to justify facts about the eigenvalues of T. For example, you might show that over a complex vector space, every operator has an eigenvalue or that any invertible map must have nonzero constant term of the minimal polynomial (for a real or Complex vector space).

Diagonalizable Operators: You should be able to define upper triangular and diagonalizable operators. You should be able to write the conditions for a basis to give an upper triangular matrix in terms of spans, and show that over a (finite-dimensional) complex vector space, there always exists a basis with respect to which an operator is upper triangular.. You should be able to determine when an operator is diagonalizable in the context of direct sums and an eigenbasis. In both cases, you should be able to determine whether an operator may be written as upper triangular or diagonal from the minimal polynomial.

Commuting Operators: You should be able to define what it means for two operators to commute, connect invariant spaces under commuting operators, and prove that two operators commute if and only if they are simultaneously diagonalizable.

Generalized Eigenvectors:

Jordan Normal Form:

3 Sets and Proofs

A set, in its essence, is just some collection of objects. That is the definition we use in discrete math, and for many purposes, though it is a naive definition (Renee Descarte and several others brought up paradoxes in this basic idea, leading the the creation of the Zermelo-Fraenkel axioms as the foundation of set theory). Most of the set theory we need is a review of Math 55, Discrete mathematics. You should be comfortable using set notation, like saying $x \in M$ for "x is an element of M", or using set-builder notation like $\{(x, x^2) \mid x \in \mathbb{R}\} \subset \mathbb{R}^2$ (*what is this set?*). You should also recognizes unions ($A \cup B$), intersections ($A \cap B$), complements (A^C), set differences ($A \backslash B$), and cartesian products ($A \times B$). This is also a good time to review the basics of proofs and generally interesting spaces, like the natural numbers $\mathbb{N}$, the integers $\mathbb{Z}$, and the real numbers $\mathbb{R}$.

1. Prove that $\left(\bigcup_{a \in A} U_a\right)^C = \bigcap_{a \in A} (U_a^C)$. (This is DeMorgan's law)
2. Define injective, surjective, and bijective. Construct a function which is injective but not surjective. Construct another that is surjective but not injective.
3. Construct a function $f : \mathbb{N} \to \mathbb{Z}$ to show that these two sets have the same cardinality (i.e. construct a bijective function).

Throughout this course, we will be using proofs. Proofs are a formal argument that gives evidence for some mathematical statement. The following is an example of such a proof presented during lecture.

Example: x^2 is even if and only if x is even.

Proof: Assume first that x is even. Then, $x = 2k$ for some integer k, and $x^2 = (2k)^2 = 4(k^2) = 2(2k^2)$. Since $m = 2k^2$ is also an integer, $x^2 = 2m$ is even. This proves the statement "If x is even, then x^2 is even".

Since we claimed an "if and only if", we also need to prove the claim "if x^2 is even, then x is even". We do this by proving the *contrapositive*, that if x is not even (odd), then x^2 is not even.

Indeed, if x is not even, $x = 2k+1$ for some integer k. Then, $x^2 = 4k^2+4k+1 = 2(2k^2+2k)+1 = 2m+1$ for $m = 2k^2 + 2k$ an integer. Therefore, x^2 also fails to be even.

The essence of our proof here is to expand on the definition in our assumption for each case, then use some other structure or calculation to show that some criteria follows from this definition. We will do this very often. One important method of proof for our course is called *proof by induction*. Mathematical induction (not electrical) is the concept that if we want to prove some condition R_n for every number $0, 1, 2, 3, 4, 5, ...$, we first show that R_0 holds (our "base case"), then show that if R_n holds, then R_{n+1} does ($R_n \Rightarrow R_{n+1}$). A great example is the following:

Example: For every non-negative integer n,

$$\sum_{i=0}^{n} i = \frac{n(n+1)}{2}$$

First, when $n = 0$, the sum is empty and so clearly both sides are 0. Next, assume that this holds for n. Then,

$$\sum_{i=0}^{n+1} i = (n+1) + \sum_{i=0}^{n} i = (n+1) + \frac{n(n+1)}{2}$$
$$= \frac{2(n+1) + n(n+1)}{2} = \frac{(n+1)(n+2)}{2}$$

so that the condition holds for $n+1$. Therefore, by the principal of mathematical induction, we have the claimed statement.

You should practice writing proofs, as they are the main mode of communicating mathematical facts.

1. Show that for all non-negative integers n,
$$\sum_{i=0}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$
2. Prove that $\mathbb{Z} \times \mathbb{Z}$ has the same cardinality as $\mathbb{Z}$.
3. Prove that there are infinitely many prime numbers. *Hint:* Assume that there are finitely many primes $p_1, ..., p_n$. Can you find a prime not in this list? If so, you have found a *contradiction* and your assumption must be false.

4 Fields

Fields are a special type of set with very nice addition and multiplication. In particular, they have structure built to generalize the properties of the real numbers. Fields are a *unital ring* in which every nonzero element has a multiplicative inverse and multiplication is commutative. Let's break down this definition. The "ring" part means that addition and multiplication are defined on the elements of the ring (where addition is commutative and associative, and multiplication is associative and distributive). This addition also has an additive identity 0 so $0+r=r$ for a ring element r, as well as an additive inverse $-r$ so $r+(-r)=0$. Next, "unital" means we have a multiplicative identity 1 so $1(r)=r$ for every element r of our field F, and the multiplicative inverse means for each r that is not zero, we have $\frac{1}{r}$ so $r\frac{1}{r}=1$. These fields are very important in abstract algebra, so it is worth looking over the full axiomatic definition a few times to understand it. We will mostly be working over the fields $\mathbb{R}$ and $\mathbb{C}$ (the complex numbers), though we will retain some abstraction. The complex numbers are particularly important because they are what is called an *algebraically closed* field, which just means that every polynomial with complex numbers as coefficients has all of its roots in the complex numbers. This fact is called the Fundamental Theorem of Algebra (a proof of it often relies on a set of theory called *complex analysis*, which is a bit beyond this course). You should recall how to multiply, add, conjugate, and divide by complex numbers.

- Which of the following are fields? $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{R}\times\mathbb{R}$ (is multiplication defined here?), $\{0,1\}$
- Consider $\mathbb{Z}_3$, which is the set $\{0,1,2\}$ where addition and multiplication are define modulo 3 (so $1+2=0$, $2+2=1$, $1+1=2$). Is this a field?
- Let $a=1+3i$ and $b=4+5i$. Compute

$$\overline{a}\frac{1}{b}$$

- Show

$$\left(\frac{-1+i\sqrt{3}}{2}\right)^3=1$$

- Show $0a=0$ for all $a\in\mathbb{F}$. *Hint:* $0=0+0$
- Show that if $1\in\mathbb{F}$ has additive inverse -1, then $(-1)a=-a$ for all $a\in\mathbb{F}$.

The "standard" structure we will use from this point on and compare many things to is $\mathbb{F}^n$, where $\mathbb{F}$ is a field. This is the set of n-tuples or lists $\{(a_1,...,a_n)\mid a_i\in\mathbb{F}\}$. Addition is defined component-wise, so

$$(a_1,...,a_n)+(b_1,...,b_n)=(a_1+b_1,...,a_n+b_n)$$

and we define a *scalar multiplication* by elements of the field: for $\alpha\in\mathbb{F}$,

$$\alpha(a_1,...,a_n)=(\alpha a_a,...,\alpha a_n)$$

1. What is $\mathbb{R}^3$?
2. Show that this multiplication is distributive, i.e. $\lambda(a+b)=\lambda a+\lambda b$ for $\lambda\in\mathbb{F}$, and $a,b\in\mathbb{F}^n$.
3. Show $1x=x$ for $x\in\mathbb{F}^n$.
4. For $x\in\mathbb{F}$, show that $x^n=x(x)(x)...(x)=0$ for some n if and only if $x=0$.

Remark: The reason we bother with this abstract definition of a field is that there are many other fields, and they can provide cases that work against our intuition. For example, we have the integers modulo a prime. "modulo" is an operation that means "the remainder when dividing by". If we are working modulo 5, we have

$$\begin{aligned} 14 + 27 = 41 = 1 + 6(5) &\equiv 1 \mod 5 \\ 2 * 4 = 8 = 3 + 1(5) &\equiv 3 \mod 5 \\ 5 = 0 + 1(5) &\equiv 0 \mod 5 \\ 4 &\equiv 4 \mod 4 \\ 4 = (-1) + 1(5) &\equiv -1 \mod 5 \end{aligned}$$

The last three cases show that working modulo 5 only requires the numbers $\{0, 1, 2, 3, 4\}$. We call the integers mod 5 by the name $\mathbb{Z}_5$. This is also a field (check the axioms, e.g. $2 * 3 \equiv 1 \mod 5$ gives a multiplicative inverse for 2) that is important in a field of math called abstract algebra.

Examples to Know: $\mathbb{Q}, \mathbb{R}, \mathbb{C}$, $\mathbb{Z}_2$.

5 Vector Spaces

5.1 Definition

A vector space is a broader abstract concept of a space with "additive structure", but the multiplicative structure is through *scalar* multiplication and so the space isn't quite as nice as just a ring. Our prototypical vector space is the $\mathbb{F}^n$ we defined above.

For any vector space, we begin with some set V and define two operations on the set. The first is addition $+ : V \times V \to V$, which is defined to be commutative and associative, have an additive identity element 0, and have for each $v \in V$ an inverse $-v$ so $v + (-v) = 0$. Next, any vector space is associated with a field $\mathbb{F}$, and we call V a vector space over the field $\mathbb{F}$. This gives the second operation, scalar multiplication $* : \mathbb{F} \times V \to V$ which is defined to be associative and distributive, while we also assume $1 * v = v$ for every vector $v \in V$.

You should note that the additive identity 0 is unique, so there cannot be two distinct vectors with the additive identity property.

In the case of $\mathbb{F}^n$, the addition was component-wise addition of vectors and the multiplication was component-wise multiplication by the field element. Therefore, all of these properties follow from the field axioms. In the future, we will actually show that many vector spaces (the finite dimensional ones) look like (are isomorphic to) our example $\mathbb{F}^n$, which is why we keep bringing this example up.

1. Using the vector space axioms, show that the set of 2×2 matrices whose elements are real numbers, $M_2(\mathbb{R})$, is a vector space.
2. Is $\mathbb{R}$ a vector space over $\mathbb{C}$? Is $\mathbb{C}$ a vector space over $\mathbb{R}$?
3. Like for fields, show $0v = 0$ for every $v \in V$ and $(-1)v = -v$ for all $v \in V$.

Examples to Know: $\mathbb{F}^n$, $\mathbb{F}^S = \{\text{functions } f : S \to \mathbb{F}\}$, the polynomials $P = \{a_0 + a_1 x + ... + a_n x^n \mid a_i \in \mathbb{F}\}$, Matrices, Sequences $S_{\mathbb{N}} = \{(a_0, a_1, ...) \mid a_i \in \mathbb{F}\}$.

Vector spaces are the foundation of our understanding of a "linear space" where we want to study linear structure. This structure seems weird and abstract at first because it is- it took many mathematicians working on affine geometry before such definitions were made. While Rene Descarte and Pierre de Fermat began identifying system solutions with plane curves as early as 1636, removing coordinates from these processes (i.e. using abstract objects) first began with Bolzano in 1804, and the modern definition of the vector space was first adopted by Giuseppe Peano in 1888 based on the work of Cayley on matrices (1857) and Grassmann's abstraction of addition and scalar multiplication (1844).

5.2 Subspaces

If we consider a vector space like $\mathbb{R}^3$, we may also notice that things which look like other vector spaces are contained in $\mathbb{R}^3$. For example, $\{(0, y, z) \mid y, z \in \mathbb{R}\}$ looks very much like $\mathbb{R}^2$. This brings us to the concept of a *subspace*. A subset U of a vector space V is a subspace if it is a vector space with the operations inherited from V. This means that, when checking to see if a subset is a subspace, we don't need to check whether the addition is associative or the multiplication is distributive, we just need to check that these operations are defined on U. In particular, we need to check that for all $u, v \in U$ and $a, b \in \mathbb{F}$, $au + bv \in U$.

1. Prove that, given $U \subset V$ satisfies this criterion, that U is a vector space.
2. Given a 2×2 matrix A, show that the solution set

$$\{x \in \mathbb{R}^2 \mid Ax = 0\}$$

is a subspace. Show that this fact is not true in general if we look at x solving $Ax = [1, 0]$.

Examples to Know: lines, planes, and other spaces in $\mathbb{R}^n$. Solutions to $Ax = 0$ for a matrix A. Polynomials of degree at-most n. Special functions, like continuous or differentiable functions. Special matrices like upper triangular or orthogonal matrices (will talk about more later in the class). Recurrence subspaces $R = \{(a_1, a_2, ...) \mid a_n = \sum_{i=1}^{m} c_i a_{n-i} \text{ for } n \geq m\} \subset S_{\mathbb{N}}$.

5.3 Sums

This definition is quite broad and gives many more subspaces than the simple form above. From Calc 3, you may remember that all planes and lines in $\mathbb{R}^3$ are subspaces. Further, if we consider the space of all functions from $[0, 1] \to \mathbb{R}$ with pointwise addition and scalar multiplication, denoted $\mathbb{R}^{[0,1]}$, the continuous functions are a subspace and the differentiable functions are a subspace of the continuous functions (and so a subspace of the original vector space).

There are many other ways to produce subspaces. For example, the intersection of subspaces is also a subspace. This is not true in general for unions: consider $A = \{(x, y) \mid y = 0\} \cup \{(x, y) \mid x = 0\}$. This is the union of the x and y axis in $\mathbb{R}^2$. Each component of the union is a subspace, but $(1, 0) + (0, 1) \notin A$ shows that A is not closed under addition.

A better way to form a larger subspace is to instead take the sum of subspaces. Given $V_1, ..., V_m \subset V$ that are subspaces,

$$V_1 + ... + V_m = \{v_1 + ... + v_m \mid v_i \in V_i \text{ for all } i\}$$

In particular, the sum is the *smallest* subspace containing each of the constituent subspaces, which is to say that if $W \subset V$ is a subspace satisfying $V_i \subset W$ for all i, then $V_1 + ... + V_m \subset W$.

1. What is $\{(x, y) \mid y = 0\} + \{(x, y) \mid x = 0\}$?
2. Prove that the intersection of subspaces is still a subspace.
3. Consider the real vector space $R = \{(a_0, a_1, a_2, ...) \mid a_i \in \mathbb{R}, \sum |a_i| < \infty\}$. Show $R_i = \{(a_0, a_1, ...) \mid a_j = 0 \text{ for } j \neq i\}$ is a subspace. Find vectors $v_i \in R_i$ so $\sum_{i=1}^{\infty} v_i \notin R$, and so we cannot extend the above definition to infinite sums without some care.
4. Prove that the union of two subspaces is a subspace if and only if one subspace contains the other.

Recall that we define $\mathbb{F}^n$ using *coordinates* $(a_1, ..., a_n)$. This is a very special case of a sum where $\mathbb{F}^n = \sum \mathbb{F}_i$ for

$$\mathbb{F}_i = \{(0, ..., 0, a_i, 0, ..., 0) \mid a_i \in \mathbb{F}\}$$

The special part is that these subspaces don't have a "large" overlap, which means that any two people looking at the same point will write the same coordinates with respect to this sum.

For example, if we add

$$\{(0,0,z,a) \mid z,a \in \mathbb{R}\} + \{(x,y,0,0) \mid x,y \in \mathbb{R}\}$$

we get all of $\mathbb{R}^4$, but we could just as well add

$$\{(x,y,0,0)\} + \{(0,y,z,a)\}$$

The way we can symbolically notice the overlap in the subspaces is that, in the first case, we may decompose any vector

$$(x,y,z,a) = (x,y,0,0) + (0,0,z,a)$$

as the sum of two vectors from the two subspaces, and there is only one possible way to do this. In the second case

$$(x,y,z,0) = (x,y-1,0,0) + (0,1,z,0) = (x,y-2,0,0) + (0,2,z,0) = ...$$

The overlapping parts of the vector space mean that we can take infinitely many different decompositions.

This leads us to the definition of a *direct sum*. Suppose $V_1, ..., V_m$ are subspaces of V. The sum $V_1 + ... + V_m$ is called a direct sum if each element $v \in V_1 + ... + V_m$ may be written uniquely as the sum $v = v_1 + ... + v_m$ for $v_i \in V_i$. In this case, we denote the sum $V_1 \oplus ... \oplus V_m$.

In particular, $U + W$ is direct if and only if $U \cap W = \{0\}$.

1. If $V_1, V_2, W \subset V$ are subspaces such that $V = V_1 \oplus W = V_2 \oplus W$, is it true that $V_1 = V_2$ (try experimenting in $\mathbb{R}^2$).?
2. Consider $\Lambda = f : \mathbb{R}^2 \to \mathbb{R}$. Let $A = \{f \mid f(x,y) = -f(y,x)\}$ and $S = \{f \mid f(x,y) = f(y,x)\}$. Show $\Lambda = A \oplus S$ (we will look at this example again much later).
3. Consider $W = \{(x,y,x,y) \in \mathbb{R}^4\}$. Find a subspace V so $W \oplus V = \mathbb{R}^4$.

6 Linear Independence and Span

Starting from any vector $v \in V$, we may immediately make a subspace by simply taking $\{cv \mid c \in \mathbb{F}\}$ (this is a line in V). Any subspace containing v must also contain this since it is closed under scalar multiplication, so this is the smallest subspace containing v. What if we take two vectors $v_1 \neq v_2$? To make a similar subspace, we must include scalar products of both vectors, as well as the sums of scalar products (all linear combinations). Thus, we define the span

$$\text{span}\{v_1, ..., v_n\} = \{\lambda_1 v_1 + ... + \lambda_n v_n \mid \lambda_i \in \mathbb{F}\}$$

to be the set of all finite linear combinations, which is again the smallest subspace containing the vectors (the word finite is used here because, if we take infinitely many vectors to build the span, we still only allow finite sums in the span). How big are these "smallest subspaces"?

We could just try counting the vectors, but we have a strange possibility that can throw a wrench in understanding their size. When $v_n \in \text{span}\{v_1, ..., v_{n-1}\}$, $\text{span}\{v_1, ..., v_n\} = \text{span}\{v_1, ..., v_{n-1}\}$. To exclude this case, we call a set of vectors $\{v_1, ..., v_n\}$ linearly independent when $v_i \notin \text{span}\{v_1, ..., v_{i-1}, v_{i+1}, ..., v_n\}$ for any i, or equivalently when

$$\lambda_1 v_1 + ... + \lambda_n v_n = 0$$

implies $\lambda_1 = ... = \lambda_n = 0$. (In the infinite-vector case, we just claim this holds for all finite subcollections of vectors).

In the case that $\{v_1, ..., v_n\}$ are linearly independent, we say their span has dimension n. Further, if $V = \text{span}\{v_1, ..., v_n\}$ and the v_i are linearly independent, we call them a *basis* of V (because we can fully

describe V by this nice span). In particular, every $w \in V$ may be written uniquely as $w = c_1v_1 + ... + c_nc_n$, and we will later call these $c_1, ..., c_n$ the coordinates with respect to this basis.

If vectors are not linearly independent, we call them linearly dependent. If $\{v_1, ..., v_m\}$ are linearly dependent, one of the vectors is in the span of the other by definition. Therefore, we may shave away unnecessary vectors until we arrive at a subcollection with the same span but which are linearly independent. In other words, every collection of vectors spanning V contains a basis. Alternately, if we start with a linearly independent collection of vectors $\{v_1, ..., v_m\}$ in V which do not span V, we can certainly add vectors from V not in their span. This means the new collection $\{v_1, ..., v_{m+1}\}$ will be linearly independent. If we keep doing this, we can fill out V and eventually obtain a basis. These are sometimes referred to as the "cutting down" and "building up" lemmas. They allow us to say things like that, in a vector space V, a linearly independent list of vectors is always at-most the length of any spanning list.

1. For a finite-dimensional vector space V, prove the "cutting down" and "building up" lemmas (in infinite dimensions, you need a tool called Zorn's lemma).
2. Prove that, in $\mathbb{F}^n$, there is no linearly independent list of vectors of length $n+1$ and no spanning list of length $n-1$.
3. If $v_1, ..., v_n$ are a basis for V, is $\{w_1, ..., w_n\}$ also a basis for $w_k = v_1 + ... + v_k$? Prove your answer.
4. For the above problem, show that the v_i are a basis if and only if the w_i are.
5. Prove $\mathbb{R}^{\mathbb{R}}$ is infinite dimensional.
6. If $\{v_1, ..., v_n\}$ and $\{w_1, ..., w_m\}$ are two lists of linearly independent vectors, is $\{v_1 + w_1, ..., v_n + w_n\}$ also linearly independent?
7. Given an arbitrary subspace $W \subset V$, prove that there is U so $W \oplus U = V$.
8. Given $U \oplus W = V$, and a basis $\{u_1, ..., u_m\}$ for U, $\{w_1, ..., w_n\}$ for W, is $\{u_1, ..., u_m, w_1, ..., w_n\}$ a basis for V? Prove your answer.

Examples to Know: The standard basis $e_1, ..., e_n$ for $\mathbb{F}^n$ ($e_1 = (1, 0, ..., 0)$, $e_2 = (0, 1, 0, ..., 0)$...). A basis for the polynomials by $\{1, x, x^2, ...\}$.

6.1 Dimension

Using the building-up and cutting down lemmas above, we know that any two bases of a finite-dimensional vector space must have the same length (verify this). We call this length the dimension of a vector space V, denoted $\dim(V)$. Using the same lemmas, we know that the dimension respects inclusion $\dim(U) \leq \dim(V)$ if $u \subset V$), and that any linearly independent set whose length is the dimension (and any spanning set whose length is the dimension) must be a basis. For example, one can check that $(1, 1)$ and $(2, 1)$ are linearly independent, and thus are a basis for $\mathbb{R}^2$ without checking span (this will appear later when we talk about linear transformations as the fact that such a transformation is injective if and only if it is surjective).

1. Check that if U and W have $U \oplus W = V$, then $\dim(U) + \dim(W) = \dim(V)$.
2. Check more generally that if $U, W \subset V$ are subspaces, then $\dim(U) + \dim(W) - \dim(U \cap W) = \dim(U + W)$.
3. Use the above two exercises to show that for U, W subspaces of V, $U \oplus W = V$ if and only if $\dim(U) + \dim(W) = \dim(V)$ and $U + W = V$.
4. Find a formula for $\dim(U + W + S)$ for subspaces $U, W, S \subset V$ (you will need at least one sum).
5. Find a basis for the set of polynomials with coefficients in $\mathbb{R}$ of degree at-most n. What is the dimension of this subspace?
6. If U and W are subspaces of $\mathbb{F}^7$, show they must have nontrivial intersection (their intersection contains a nonzero element).

Examples to Know: You should know how to prove all of the above mentioned facts, and connect some common subspaces (of $\mathbb{F}^n$) to pictures via their dimension. The polynomial subspaces are also important here.

7 Linear Maps

A linear map $T : V \to W$ is a function between vector spaces over a field $\mathbb{F}$ which respects their operations, so $T(au + bv) = aT(u) + bT(v)$ for $a, b \in \mathbb{F}$ and $u, v \in V$. In particular, the set of linear maps from V to W, denoted $L(V, W)$, is itself a vector space when we define pointwise addition and scalar multiplication for these functions. Some common examples are

- The map $T : V \to W$ which maps all vectors to $0 \in W$
- The identity $T : V \to V$ so $T(v) = v$
- Matrices $M(n)$ acting on $\mathbb{R}^n$
- Differentiation and integration
- Multiplication by a field element (scalar multiplication)
- $T : S_{\mathbb{R}} \to S_{\mathbb{R}}$ which takes $T(a_1, a_2, ...) = (a_2, a_3, ...)$ (this is the Bernoulli Shift).

We can add even more structure to $L(V, W)$ when we define the product of $T \in L(U, V)$ and $S \in L(V, W)$ to be the composition $ST = S \circ T$, since this is associative and distributive.

1. Prove that if $T \in L(V, W)$, then $T(0) = 0$.
2. Is the map $T : \mathbb{P} \to \mathbb{R}^2$ defined by
$$T(p(x)) = (p(0), \int_0^1 p(x)dx, p'(0))$$
linear?
3. Let $\{v_1, ..., v_n\}$ be a basis for V and $\{w_1, ..., w_m\}$ be a basis for W with $m \geq n$. Construct a linear map $T : V \to W$ which takes $T(v_i) = w_i$ for $i = 1, ..., n$. *Hint:* If this map exists, what is $T(av_1 + bv_2)$?
4. Consider the map $\phi : \mathbb{R}^2 \to \mathbb{R}$ given by $\phi(x, y) = x + 1$. Is it homogeneous? Is it additive? Answer the same questions for $\psi : \mathbb{R}^2 \to \mathbb{R}$ given by $\psi(x, y) = \sqrt{x^2 + y^2}$. What does this show you about these two properties we use to define linearity?
5. Is the map $T : \mathbb{P} \to \mathbb{P}$ given by $T(p(x)) = q(p(x))$ for some $q \in \mathbb{P}$ a linear map?
6. Suppose V is a finite dimensional vector space and $T \in L(V)$ is such that $ST = TS$ for every $S \in L(V)$. Prove T is a scalar multiple of the identity (a *homotethy*). *Hint:* Consider the two cases where $\dim(V) = 1$ and $\dim(V) > 1$. In the first case, show every linear map must be a homotethy. In the second case, show that if $T \neq \alpha I$, then there always exists S so $ST \neq TS$.

As with many functions, we consider the case of injective and surjective linear maps. In particular, the linear structure means that we generate some interesting subspaces when looking into these.

First, for any $w \in W$ and $T \in L(V, W)$, and any v_0 so $T(v_0) = w$,

$$\{v \in V \mid T(v) = w\} = v_0 + \{v \in V \mid T(v) = 0\}$$

so that T is injective if and only if second set is precisely $\{0\}$. We give this set a name, the *null space* or the *kernel* $Null(T) = \{v \in V \mid T(v) = 0\}$. You may prove that the null space is actually a subspace, and we will talk a lot more about it when it comes to quotient spaces.

Similarly, the image or range of the linear map is also a subspace, and we denote it $Im(T) = \{Tv \mid v \in V\}$. As with normal functions, T is surjective if and only if this image is all of W.

However, it is useful to understand these two subspaces because of the Rank-Nullity Theorem: if V is a finite-dimensional vector space and $T \in L(V, W)$, then

$$\dim(V) = \dim(Null(T)) + \dim(Im(T))$$

(i.e. the dimension is the nullity plus the rank) This is an important proof to know, and so we sketch it here. Begin by establishing a basis $\{u_1, ..., u_m\}$ for $Null(T)$, then extend it to a basis for V $\{u_1, ..., u_m, v_1, ..., v_n\}$ Notice that $Im(T)$ is spanned by $\{T(u_1), ..., T(u_m), T(v_1), ..., T(v_n)\}$, but of course the first m vectors here are 0 so it is spanned by $\{T(v_1), ..., T(v_n)\}$. Furthermore, these vectors are linearly independent: if $\alpha_1 T(V_1) + ... + \alpha_n T(V_n) = 0$, then

$$T(\sum \alpha_i v_i) = 0$$

and so $\sum \alpha_i v_i \in Null(T)$. By the basis assumption, we must then have that $\sum \alpha_i v_i = 0$, and so $\alpha_i = 0$ since the v_i are linearly independent. Therefore, $\dim(Im(T)) = n$, $\dim(Null(T)) = m$, and $\dim(V) = m + n$ as we claimed.

Some remarkable results come from this theorem. For example, $T \in L(V)$ is injective if and only if it is surjective (and so it is bijective and invertible if and only if it is injective). If $T \in L(V, W)$ and $\dim(V) > \dim(W)$, then T *must* fail to be injective. If instead $\dim(V) < \dim(W)$, it must fail to be surjective. This is also how we prove that a homogeneous system of linear equations has nonzero solutions!

1. Is the derivative from $\mathbb{P} \to \mathbb{P}$ injective? Is it surjective?
2. Prove the facts claimed in the last paragraph above.
3. Let V have odd dimension. Show there is no $T \in L(V)$ so $Im(T) = Null(T)$. If W has even dimension, give an example of $S \in L(W)$ so $Im(S) = Null(S)$.
4. Show that the set of injective linear maps is not a subspace of $L(V, W)$.
5. Prove the converse of some of the facts claimed in the last paragraph: if $\dim(V) \leq \dim(W)$, then there exists an injective linear map from V to W. Similarly, if $\dim(V) \geq \dim(W)$, there exists a surjective map.
6. Let V be infinite-dimensional. Show that for any $T \in L(V, W)$, either $Im(T)$ or $Null(T)$ must be infinite-dimensional.
7. Given $T : V \to W$ and $S : W \to U$, show that if ST is injective, T must be injective. Furthermore, show

   $$Null(ST) = T^{-1}(Null(S))$$

   and

   $$\dim(Null(ST)) \leq \dim(Null(S)) + \dim(Null(T))$$

   with equality if and only if $Null(S) \subset Im(T)$ (Hint: for $T : V \to W$ and V finite-dimensional, $\dim(T^{-1}(U)) \leq \dim(Null(T)) + \dim(U \cap Im(T))$).
8. (Sided inverses) Let $T : V \to W$ be linear. Show that T is surjective if and only if there exists $S \in L(W, V)$ so TS is the identity. Show that T is injective if and only if there is $S \in L(W, V)$ so ST is the identity.
9. There are a ton of problems here and a lot more in Axler's chapter on this- pick 5 from his chapter that don't repeat information from the above (29 is cool)

Examples to Know: There aren't many affixed examples here, but you should be able to come up with various simple examples of injective, non-injective, surjective, non-surjective, linear, and non-linear maps. Projection maps and inclusion maps are often quite helpful.

7.1 Matrices

When V and W are both finite-dimensional vector spaces, it is easy to represent a linear map $T : V \to W$ in an information-dense format called a matrix. In particular, we establish a pair of bases $\{v_1, ..., v_m\}$ for V and $\{w_1, ..., w_n\}$ for W. Then, we may write every $v \in V$ as $v = \sum_{i=1}^{m} a_i(v)v_i$ for some scalars $a_i(v) \in \mathbb{F}$, and we may similarly decompose W. In this situation, we can give v the "name" or coordinates $(a_1, ..., a_m)$.

Let $T(v_i) = \sum_{j=1}^{n} b_{i,j}w_j$. Notice that for any $v = (a_1, ..., a_n)$,

$$\begin{aligned} T(v) = \sum_{i=1}^{m} a_i T(v_i) &= \sum_{i=1}^{m}\sum_{j=1}^{n} a_i b_{i,j} w_j \\ &= \sum_{j=1}^{n}[\sum_{i=1}^{m} a_i b_{i,j}]w_j \end{aligned}$$

so in coordinates on W, this is $([\sum_{i=1}^{m} a_i b_{i,1}], ..., [\sum_{i=1}^{m} a_i b_{i,m}])$. You may recall from a previous linear algebra class that this is precisely the vector generated by

$$[b_{i,j}] \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix}$$

and so we can fully encode the action of T by this matrix $[b_{i,j}]$ for scalars $b_{i,j}$ (what size is this matrix?). We denote this matrix as $\mathcal{M}(T)$. Notice that the sum and scalar product of matrices is the same as that of the linear operations:

$$\mathcal{M}(\lambda T + S) = \lambda \mathcal{M}(T) + \mathcal{M}(S)$$

whereas matrix multiplication is precisely composition

$$\mathcal{M}(S)\mathcal{M}(T) = \mathcal{M}(ST)$$

(justifying why we call it the product of linear maps). You should be able to prove these two facts by evaluating on vectors. For example, let $\mathcal{M}(S) = [a_{i,j}]$ be a $p \times m$ matrix (why do we need m columns?). Then,

$$[\mathcal{M}(S)\mathcal{M}(T)]_{j,k} = \sum_{r=1}^{m} a_{j,r} b_{r,k}$$

and so

$$\mathcal{M}(S)\mathcal{M}(T) \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix} = \begin{bmatrix} \sum_{k=1}^{n}\sum_{r=1}^{n} a_{1,r} b_{r,k} c_k \\ \vdots \\ \sum_{k=1}^{n}\sum_{r=1}^{n} a_{p,r} b_{r,k} c_k \end{bmatrix}$$

and you should get the same when looking at the composition.

This matrix depends quite heavily on the choice of bases involved. If we call the basis $\{v_1, ..., v_m\}$ for V by $\mathcal{A}$ and $\{w_1, ..., w_n\}$ for W by $\mathcal{B}$, then we may instead write $\mathcal{M}_{\mathcal{A}}^{\mathcal{B}}(T)$. We write the domain in the bottom so this can look like fractions canceling when we keep track of bases:

$$\mathcal{M}_{\mathcal{B}}^{\mathcal{C}}(S)\mathcal{M}_{\mathcal{A}}^{\mathcal{B}}(T) = \mathcal{M}_{\mathcal{A}}^{\mathcal{C}}(ST)$$

In general, to find the matrix for T under the bases $\mathcal{A}$ and $\mathcal{B}$, we follow the computations we did original to notice that the kth column of $\mathcal{M}_{\mathcal{A}}^{\mathcal{B}}(T)$ is

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

where

$$T(v_k) = \sum_{j=1}^{n} c_j w_j$$

(the coordinates of this image under $\mathcal{B}$).

One of our next big topics will look at how we can move between these coordinate representations and bases, and what that means for these matrices.

1. Let $T : \mathbb{P}_3 \to \mathbb{P}_2$ be given by $T(p(x)) = p'(x) + p''(x)$. Find bases for these two spaces and the matrix representation of T under these bases.

2. Consider $\mathbb{R}^2$ under the basis e_1 and e_2, and let T be given by the matrix

$$\begin{matrix} 1 & 1 \\ 0 & 1 \end{matrix}$$

Under the alternative basis

$$\{\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}\}$$

consider the matrix

$$\begin{bmatrix} 5/3 & 4/3 \\ -1/3 & 1/3 \end{bmatrix}$$

Does this matrix represent T?

8 Isomorphisms

How do we know when two vector spaces are equal, and what should that mean? A copy of $\mathbb{R}^2$ sits in $\mathbb{R}^3$: $\{(x, y, 0) \mid x, y \in \mathbb{R}\}$ looks identical to $\mathbb{R}^2$ but is not strictly "equal".

To answer this, we recall that two sets A, B with no extra structure on them have the same size if and only if there exists a bijection $f : A \to B$ between them (a function which is both injective and surjective, which is the same as being invertible, which means to have $f^{-1} : B \to A$ so $f \circ f^{-1} = Id_B$ and $f^{-1} \circ f = Id_A$). Since we are adding some additive structure and some scalar multiplication structure, equality would mean we just want these two operations to look the same on either side of the bijection, so $f(\lambda v + w) = \lambda f(v) + f(w)$. However, this is just a linear map!

Therefore, we call a linear map $T \in L(V, W)$ an isomorphism if it is invertible, or equivalently if it is bijective. In this case, the inverse is also linear. We call two vector spaces V and W isomorphic if there exists a linear map between them.

Checking an isomorphism is easier than normal in finite-dimensional vector spaces of the same dimension, since the Rank-Nullity theorem implies that a map T is invertible iff. it is injective iff. it is surjective. Recall that a linear map $T : V \to W$ is injective if and only if there exists $S \in L(W, V)$ so $TS = I$. Similarly, T is surjective if and only if there exists $S : W \to V$ so $ST = I$. Therefore, the previous claim from the Rank Nullity theorem says that when V and W have the same dimension, a left inverse exists if and only if a right inverse exists, in which case this is a two-sided inverse T^{-1}.

Notice further that invertible linear maps must take bases to bases: if $\{v_1, ..., v_n\}$ is a basis for V and $T : V \to W$ is invertible and linear, $\{T(v_1), ..., T(v_n)\}$ is a basis for W. To see this, notice that the map being injective implies the $T(v_i)$ are linearly independent, and the map being surjective implies they span W. In particular, V and W must have the same dimension.

Given any V and W of the same dimension, we can also find a pair of bases $\{v_1, ..., v_n\}$ and $\{w_1, ..., w_n\}$ and construct a map so $T(v_i) = w_i$. It can be shown that this map is then injective and surjective. This means that two finite-dimensional vector spaces are isomorphic if and only if they have the same dimension

(so every n-dimensional vector space over $\mathbb{F}$ is isomorphic to $\mathbb{F}^n$). For example, $L(V,W)$ is isomorphic to $\mathbb{F}^{\dim(V)\dim(W)}$, where the canonical isomorphism is by using the matrix of a linear transformation.

This revelation about bases and dimension tells us that every invertible transformation looks like a change-of-basis and every change-of-basis is just an isomorphism.

1. Given that $TU \to V$ and $S : V \to W$ are isomorphisms, check that TS is.
2. If $TS : U \to W$ is an isomorphism of finite dimensional vector spaces, are $T : V \to W$ and $S : U \to V$ also are isomorphisms.
3. Suppose V is finite-dimensional and $S, T \in L(V,W)$. Show $range(S) = range(T)$ if and only if there exists invertible $E \in L(V)$ so $S = TE$.
4. Show that V is isomorphic to $L(\mathbb{F}, V)$.

9 Products, Direct Sums, Quotient Spaces

We have found ways to observe vector spaces, to differentiate them, and to relate some subspaces to other vector spaces. We now look at ways to produce new vector spaces and create such relationships.

The first and easiest case is that of the product. Given two vector spaces V and W over the same field $\mathbb{F}$, define

$$V \times W = \{(v,w) \mid v \in V, w \in W\}$$

with the component-wise operations $(v_1, w_1) + a(v_2, w_2) = (v_1 + av_2, w_1 + aw_2)$. This is a new vector space, with $\dim(V \times W) = \dim(V) + \dim(W)$.

One interesting application of this product is when V_1, V_2 are subspaces of a vector space V. Then, if we have $V_1 \cap V_2 = \{0\}$, the direct sum $V_1 \oplus V_2$ is isomorphic to this product $V_1 \times V_2$. This is seen in an easy way: for arbitrary subspaces W_1, W_2, the sum $W_1 + W_2$ is direct if and only if the map $T(w_1, w_2) : W_1 \times W_2 \to W_1 + W_2$ given by $T(w_1, w_2) = w_1 + w_2$ is injective.

While it is easy to build up like this, there is a slightly more complicated way to "shrink" vector spaces. It comes from the idea of a linear map: assume we have a vector space V and subspace U. Then, if $U = \ker(T)$ for a linear map $T : V \to W$, the image of T "looks like" pushing everything in U to 0, and keeping the rest of V up to this identification. If we view vector space products as "repeated addition/multiplication" , this amounts to "repeated subtraction/division". This is how we begin to construct quotient spaces.

For this definition, let V be a vector space and $U \subset V$ a subspace. For $v \in V$, define $v + U = \{v + u \mid u \in U\}$, which is called the translate of U. Notice that for $v, w \in V$, we may have two possibilities. First, $v - w \in U$, in which case $v + U = w + U$. Second, $v - w \notin U$, in which case $v + U \cap w + U = \emptyset$.

We define the quotient space V/U to be the set of translates

$$\{v + U \mid v \in V\}$$

By our observation above, these are distinct subsets partitioning V. If we further define addition and scalar multiplication

$$(v + U) + (w + U) = (v + w) + U$$
$$\lambda(v + U) = \lambda v + U$$

then V/U is a vector space.

Given arbitrary V, U, we may now define a map $\pi : V \to V/U$ by $\pi(v) = v + U$. Notice that the kernel of this map is precisely U, so *every subspace is the kernel of a linear map.*

Given an arbitrary linear map $T : V \to W$, we can also construct a map $\tilde{T} : V/\ker(T) \to W$ given by

$$\tilde{T}(v + \ker(T)) = T(v)$$

which is well-defined because for any $w \in v + U$, $w = v + u$ for some $u \in U$ and $T(w) = T(v) + T(u) = T(v)$. This new map $\tilde{T}$ is also injective with the same image as T, and so it gives an isomorphism $V/\ker(T) \simeq Im(T)$. Hence, every quotient space is isomorphic to the image of a linear map. This is a big concept that will appear later in your math journey as well, and is commonly called the first isomorphism theorem.

1. Given a basis $\{v_1, ..., v_m\}$ for V and $\{w_1, ..., w_n\}$ for W, find a basis for $V \times W$.
2. Let $T : V \to V$ and $S : W \to W$ be given, with matrix representations A and B, let $T \times S(v, w) = (T(v), S(w))$. What is the matrix representation for $T \times S$?
3. Show that $L(V, W_1 \times W_2) \simeq L(V, W_1) \times L(V, W_2)$ as vector space.
4. Prove that $V^m \simeq L(\mathbb{F}^m, V)$ as vector spaces.
5. Prove that the translate addition and scalar multiplication are well defined: i.e. show that if $v_1 + U = v_2 + U$, then $\lambda(v_1 + U) = \lambda(v_2 + U)$ and if $w_1 + U = w_2 + U$ also, then $(v_1 + U) + (w_1 + U) = (v_2 + U) + (w_2 + U)$.
6. Let $U \subset V$ be a subspace. If $\{v_1 + U, ..., v_m + U\}$ is a basis for V/U and $\{u_1, ..., u_n\}$ for U, show $\{v_1, ..., v_m, u_1, ..., u_n\}$ is a basis for V.
7. Let $U \subset V$ be a subspaces and V/U be finite-dimensional. Prove $V \simeq U \times V/U$. *Hint:* You cannot use the above problem because U may be infinite-dimensional. However, you can still use a basis of V/U to construct a map $U \times V/U \to V$ which is injective.
8. Let $U \subset V$ be a subspace so V/U is finite-dimensional. Show there exists W so $W \oplus U = V$.

Outside of Axler's coverage, the Professor also brings up the second and third isomorphism theorems. These both correspond to cases where we obtain useful isomorphisms by doing further calculations with quotient spaces.

The second isomorphism theorem claims that for U, W subspaces of V, there is a natural isomorphism

$$\phi : U/(U \cap W) \to (U + W)/W$$

which shows $\dim(U) - \dim(U \cap W) = \dim(U + W) - \dim(W)$, a formula we already know! The proof is simple: we take a linear map $T : U \to (U + W)/W$ by $T(u) = u + W$. Since $U \cap W = \ker(T)$, we use the first isomorphism theorem to obtain an isomorphism

$$\tilde{T} : U/(U \cap W) \to (U + W)/W$$

The third isomorphism theorem similarly claims that when $W \subset U \subset V$ there is a natural isomorphism

$$\phi : (V/W)/(U/W) \to V/U$$

The proof is similar to the above. Since $W \subset U$, the map $T : V/W \to V/U$ given by $T(v + W) = v + U$ is well-defined and its kernel is U/W, so applying the first isomorphism theorem gives the claim.

10 Duality

Duality is a very important concept moving forward in math, and it all stems from a simple fact: $L(V, \mathbb{F}) \simeq F$ as vector fields when V is finite-dimensional. Because of this special isomorphism, we define $V' = L(V, \mathbb{F})$ to be the dual of V, which is also the set of linear functionals on V (linear maps from V to its base field).

Given a (finite) basis $\{v_1, ..., v_m\}$ for V, there is a canonical dual basis $\{\phi_1, ..., \phi_m\}$ for V' where the ϕ_i are functions so

$$\phi_i(v_j) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Notice that if $v = \sum c_i v_i$, then $c_i = \phi_i(v)$, and we may write $v = \sum \phi_i(v) v_i$.

We can also lift maps to dual maps: given $T \in L(V, W)$, there exists a dual map (sometimes called the pullback) $T' \in L(W', V')$ defined by $T'(\phi) = \phi \circ T$. Dual maps have the normal properties: $(S+T)' = S' + T'$,

$(\lambda T)' = \lambda(T')$, but this dual actually flips the order of composition (because it flips the direction of the map in a sense): $(ST)' = T'S'$.

This should look very familiar. To describe it more concretely, let us consider the matrices of the map T. Give a basis $\{v_1, ..., v_m\}$ to V and $\{w_1, ..., w_n\}$ to W. Denote the dual bases using v_i' and w_j'. Finally, let T have matrix A under these bases. Then,

$$T'(w_j') = \sum_{r=1}^{n} c_{r,j} v_i'$$

Applying both sides to v_k,

$$w_j'(T(v_k)) = w_j'(\sum A_{r,k} w_r) = A_{j,k}$$
$$\sum_{r=1}^{n} c_{r,j} v_i'(v_k) = c_{k,j}$$

Therefore, T' has the matrix A^T, and the dual map is precisely the transpose map when the vector spaces are finite dimensional! Recall that we know facts like A is injective if and only if A^T is surjective (counting pivots), and so now we should hope to prove similar facts for arbitrary T and T' where we may have infinite-dimensional spaces and not be able to use pivots and matrices.

For this purpose, we define the annihilator of a subset $U \subset V$, denoted U^0, to be

$$U^0 = \{\phi \in V' \mid \phi(u) = 0 \text{ for all } u \in U\}$$

Notice that the annihilator is a subspace precisely when U is, and when V is finite-dimensional, $\dim(U^0) = \dim(V) - \dim(U)$ (build up a basis from U and then use its dual basis; alternately, use rank-nullity on the inclusion of U into V).

Recall that for matrices in $\mathbb{R}^n$, $Null(A) = (Col(A^T))^{\perp}$. We haven't used inner products yet in this course, but that helps make the following look more familiar.

Suppose V and W are finite-dimensional and $T \in L(V, W)$. Then, $\ker(T') = (Im(T))^0$ and so $\dim(\ker(T')) = \dim(\ker(T)) + \dim(W) - \dim(V)$. We have similarly that $Im(T') = (\ker(T))^0$, which dimensionally says that $\dim(Im(T)) = \dim(Im(T'))$, or that the column rank of a matrix is equal to the row rank. In particular, these two results imply that T is surjective if and only if T' is injective, whereas T is injective if and only if T' is surjective.

1. Prove all of the results in the previous paragraph (you should prove the set equalities directly using the definition of the sets and inclusion, then build up statements about dimension).
2. Show that any nonzero linear functional is surjective.
3. Given vector spaces V_1, V_2, what is $(V_1 \times V_2)'$?
4. Consider the basis $1, x, ..., x^m$ for $\mathbb{P}_m$. What is its dual basis?
5. Let V be a finite-dimensional vector space and consider $(V')' = V''$, the second dual. For each $v \in V$, define ϕ_v to be the map $\phi_v(\psi) = \psi(v)$. Use the function $F(v) = \phi_v$ to show $V \simeq V''$.
6. Show that $W^0 \subset U^0$ if and only if $U \subset W$. If V is finite dimensional, show also that $(U+W)^0 = U^0 \cap W^0$
7. Show that $(U^0)^0 = U$ under the association $V \simeq V''$ given above.
8. If V, W are finite-dimensional and $T \in L(V, W)$ has $\ker(T') = \text{span}(\phi)$ for some $\phi \in W'$, then $Im(T) = \ker(\phi)$.

11 Invariant Spaces, Eigenspaces

We focus on the set of operators $L(V)$. An invariant subspace $U \subset V$ is one where $T(U) \subset U$ (equality is not necessary). When span(v) is invariant (for $v \notin \ker T$), $Tv = \lambda v$, and v is called an eigenvector with eigenvalue λ. A nonzero value λ is then an eigenvalue precisely when $T - \lambda I$ fails to be injective (or fails to be surjective, due to Rank-Nullity). We call the set of all eigenvectors corresponding to a specific eigenvalue the eigenspace for that eigenvalue, denoted $E(\lambda, T) = \ker(T - \lambda I)$.

Eigenvectors for different eigenvalues are naturally linearly independent: let $\{v_1, ..., v_n\}$ be a list of eigenvectors with nonequal eigenvalues $\{\lambda_1, ..., \lambda_n\}$. Then,

$$\begin{aligned} &\alpha_1 v_1 + ... + \alpha_n v_n = 0 \qquad \text{Apply } T - \lambda_1 I \\ &\Rightarrow \\ &\alpha_2(\lambda_2 - \lambda_1)v_2 + ... + \alpha_n(\lambda_n - \lambda_1)v_n = 0 \end{aligned}$$

If we repeat, applying $T - \lambda_i I$ for $i = 2...n-1$, we arrive at $\alpha_n \prod_{i=1}^{n-1}(\lambda_n - \lambda_i)v_n = 0$, so we must have $\alpha_n = 0$. We then move back one step to notice next that $\alpha_{n-1} = 0$, and eventually that all the coefficients are 0.

1. Show that the sum and intersection of T-invariant subspaces are invariant. Show that any subspace of $\ker T$ is invariant, as is any subspace containing the image of T.

2. What are the eigenvalues of $T(x, y) = (x + 2y, 2x + 2y)$? What are the associated eigenvectors?

3. What are the eigenvalues and eigenvectors of the projection $\pi : \mathbb{R}^4 \to \mathbb{R}^4$ $\pi(x, y, z, a) = (0, y, z, 0)$.

   In general, if $V = U \oplus W$, what are the eigenvalues and eigenvectors of the projection into U, $\pi(u, w) = (u, 0)$?

4. Let V be finite-dimensional. Show that the eigenvalues of T and T' (the dual map) are the same (use injectivity).

   For $S \in L(V)$ invertible, show that T and $S^{-1}TS$ also have the same eigenvalues.

   If T is invertible and λ is an eigenvalue, show λ^{-1} is an eigenvalue of T^{-1}

   For arbitrary $S \in L(V)$, show ST and TS have the same eigenvalues.

5. Find a linear map with no eigenvalues *Hint:* Try looking at $\mathbb{F}^\infty$.

Remark: You should be comfortable showing that eigenvectors for different eigenvalues are linearly independent (like the above), as well as computing eigenvalues and eigenvectors of matrices (from Math 54).

Therefore, we may only have $\dim(V)$ distinct eigenvalues. How do we find them? In a previous class, you would immediately jump to the Cayley polynomial $\det(M(T) - xI)$ and find the zeroes (since the eigenvalues of T and the eigenvalues of its matrix are the same). Here, we want to build up the understanding of such a polynomial a bit further first.

12 Minimal Polynomial

Our general idea comes from factoring polynomials. Recall the Fundamental Theorem of Algebra: a degree-n polynomial has n roots in $\mathbb{C}$. In particular, it may be factored as $a\prod(z - r_i)$. We want to find some relationship between T and constants so that $(T - \lambda I)v = 0$. What we do know is that we can make a "polynomial" zero with T: since $v, Tv, ..., T^n v$ are $n + 1$ vectors in a n-dimensional space,

$$a_n T^n v + ... + a_0 v = 0$$

for some constants a_i. If we could "factor" this like a polynomial, we would arrive at

$$[a\prod_{i=1}^{n}(T - \lambda_i I)]v = 0$$

for some λ_i. Therefore, we may expect that a "factor" has $(T - \lambda_i I)v = 0$, and we have an eigenvalue and eigenvector (this is not quite true, but is close).

To make this more precise, let $p(z) = \sum_{i=0}^{m} a_i z^i$ be a polynomial. Then, we define the operator

$$p(T) = a_0 I + a_1 T + ... + a_m T^m$$

where T^m is applying T m times. Then, $(pq)(T) = p(T)q(T) = q(T)p(T)$ shows that this factoring makes sense. Furthermore, the null space and range of $p(T)$ are both T-invariant subspaces.

We can again turn the linear dependence relationship

$$a_n T^n v + ... + a_0 v = 0$$

into a polynomial $p(z) = \sum a_i z^i$, with factored form $p(z) = a \prod (z - \lambda_i)$. Then, using the product property of the operators,

$$0 = p(T)v = [a \prod (T - \lambda_i I)]v$$

Now, we can find *some* eigenvector out of this, with an eigenvalue which is one of the λ_i. Indeed, since $a \neq 0$,

$$0 = [\prod (T - \lambda_i I)]v = (T - \lambda_1 I) \prod_{i=2}^{n} (T - \lambda_i I)v = (T - \lambda_1 I)q(T)v$$

If $q(T)v \neq 0$, then this is the eigenvector with eigenvalue λ_1. If it is zero, we have

$$0 = \prod_{i=2}^{n} (T - \lambda_i I)v$$

and repeat the test. Therefore, we have some complex eigenvalue.

We may push this much farther. It is actually true that for a finite-dimensional vector space V over a field $\mathbb{F}$ and $T \in L(V)$, there is a unique monic polynomial $p \in \mathbb{P}(\mathbb{F})$ of degree at-most $\dim(V)$ so $p(T) = 0$ (and so the process above works for all vectors v). The proof is by induction, where we use a polynomial of the above form to reduce the dimension we are focused on.

The nonzero monic polynomial of least degree which satisfies $p(T) = 0$ is called the minimal polynomial. The eigenvalues are then the zeroes of this minimal polynomial. Since we know there is a polynomial of degree at-most $\dim(V)$ satisfying this, we know that the minimal polynomial must exist and must have degree at-most $\dim(V)$.

If V is a complex vector space, any polynomial $p(z)$ which has $p(T) = 0$ may be factored, and to make it minimal, we shouldn't have any repeated roots. In particular, the minimal polynomial is

$$(z - \lambda_1)...(z - \lambda_m)$$

for distinct eigenvalues $\lambda_1, ..., \lambda_m$ of T (using the above and the Fundamental Theorem of Algebra).

Furthermore, any $q(T)$ so $q(T) = 0$ must be a polynomial multiple of the minimal polynomial (by polynomial division). Relationships between operators also become relationships between these polynomials. For example, the minimal polynomial of T is a polynomial multiple of the minimal polynomial of $T|_U$.

Lastly, notice that the constant term of the minimal polynomial is the *product of the roots*, and hence *the product of the eigenvalues* (the determinant, as we will discuss later). Therefore, since T is invertible if and only if its eigenvalues are all nonzero, it is invertible if and only if the constant term of the minimal polynomial is nonzero.

1. Let $T \in L(V)$. Show that $\lambda \in \mathbb{R}$, $\lambda > 0$ is an eigenvalue of T^2 if and only if $\sqrt{\lambda}$ or $-\sqrt{\lambda}$ is an eigenvalue of T.

2. Extend the above as follows. Let V be a complex vector space, p a polynomial, and $\alpha \in \mathbb{C}$. Prove that α is an eigenvalue of $p(T)$ if and only if $\alpha = p(\lambda)$ for λ some eigenvalue of T.

3. Compute the minimal polynomial of

$$T = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$
$$S(x_1, x_2, x_3) = (x_1, 2x_2, 3x_3)$$

(we will give a better algorithm to compute this later in the course).

4. Prove that T is invertible if and only if $I \in \text{span}(T, T^2, ..., T^{\dim(V)})$ *Hint:* use the statement about invertibility and the constant of the minimal polynomial.

5. Given T with minimal polynomial $p(z)$, what is the minimal polynomial of the dual map T'? If T is invertible, what is the minimal polynomial of T^{-1}?

Remark: You should be comfortable finding a polynomial T so $p(T) = 0$, as well as reducing this polynomial (using factors) to the minimal polynomial. You should also be comfortable moving between minimal polynomials when we apply some operation to T, such as taking ST, T^2, or $T|_U$.

13 Diagonalizability

13.1 Upper-Triangular Matrices

In this section, we care about the matrix for an operator $T \in L(V)$, so we will assume V is finite-dimensional.

Recall that upper-triangular matrices are those with $a_{ij} = 0$ if $i > j$ (i.e. zeroes below the diagonal). They are useful when it comes to eigenvalue calculations because their eigenvalues are the numbers on the diagonal. To see this fact, notice that

$$M(T - \lambda I) = \begin{bmatrix} \lambda_1 - \lambda & * & * \\ 0 & \ddots & * \\ 0 & 0 & \lambda_n - \lambda \end{bmatrix}$$

so that this fails to have the full number of pivots precisely when $\lambda = \lambda_i$ for some i.

Another way to view an upper triangular matrix is, if we consider $M(T)$ written with respect to the basis $\{v_1, ..., v_n\}$, that $T(v_i) \in \text{span}\{v_1, ..., v_{i-1}, v_i\}$, so that for each i, this is an invariant subspace. In this light, when we can pick out the eigenvalues of a linear transform T, we can force such a structure. To be precise, if V is a finite-dimensional complex vector space, then there is some basis under which T is upper triangular. The proof is by induction. The one-dimensional case is immediate by picking an eigenvector. In general, we start with eigenvalue λ of T (guaranteed since we have a complex vector space) and set $U = Im(T - \lambda I)$. This dimension smaller than that of V since $T - \lambda I$ fails to be injective. Therefore, $T|_U$ may be written as upper-triangular under basis $\{u_1, ..., u_m\}$. If we extend this to $\{u_1, ..., u_m, v_{m+1}, ..., v_n\}$ for V, we notice that

$$Tv_k = (T - \lambda I)v_k + \lambda v_k \in \text{span}\{u_1, ..., u_m, v_k\}$$

for each k, and so T is upper triangular with respect to this basis.

More generally, on a finite-dimensional vector space V (not necessarily complex), T may be written as an upper triangular matrix if and only if the minimal polynomial of T is $(z - \lambda_1)...(z - \lambda_m)$ for some $\lambda_1, ..., \lambda_m \in \mathbb{F}$ (this proof is similar to the above and is a good exercise).

1. Show that the product of two upper-triangular matrices is upper-triangular.

2. Is it true that if A^2 is an upper-triangular matrix, that A also is (try viewing this with matrices and also with the eigenvalue comment in the paragraph above).

3. Let V be a complex vector space, with $1 \leq k \leq \dim(V)$. Does T have a k-dimensional invariant subspace?

4. If T has an upper triangular matrix (on finite-dimensional V) and U is T-invariant, show that the induced map $\pi \circ \tilde{T} : V/U \to V/U$ also has an upper-triangular matrix on some basis of V/U. Show also that the dual map T' has an upper-triangular matrix.

Remark: You should be able to conjure up upper triangular matrices on command, identify upper triangular matrices, and associate them to special bases. You should also be able to use their special form to quickly solve linear systems and compute eigenvalues.

13.2 Diagonal Matrices

Our real goal here is to be even more precise, that over a complex vector space T may be written with a highly specific structure (we will show later that this is the Jordan Normal Form).

We start by looking at linear transformations that may be written as diagonal matrices, since these are quite easy to work with. A diagonal matrix is a matrix (a_{ij}) where $a_{ij} = 0$ if $i \neq j$. An operator on V is called diagonalizable if it has a diagonal matrix with respect to some basis of V. Notice that if the basis is $\{v_1, ..., v_n\}$, this implies each v_i is an eigenvector with eigenvalue a_{ii}. This is to say that we have a basis of eigenvectors.

Recall that $\dim(E(\lambda, T))$ is called the geometric multiplicity of the eigenvalue λ. As we noted before, the eigenspaces $E(\lambda, T)$ have trivial intersections, so

$$E(\lambda_1, T) + ... + E(\lambda_n, T)$$

is a direct sum. In particular, this implies

$$\dim(E(\lambda_1, T)) + ... + \dim(E(\lambda_n, T)) \leq \dim(V)$$

On each eigenspace, $T|_{E(\lambda_i, T)}$ may be written as diagonal. Therefore, if T is diagonal, we have equality

$$\dim(E(\lambda_1, T)) + ... + \dim(E(\lambda_n, T)) = \dim(V)$$

and if we have equality, we may use the direct sum to generate a basis under which T is diagonal (T is diagonalizable if and only if we have equality here).

A good example using this is that the matrix

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

is not diagonalizable because it only has eigenvalue 1 and that eigenspace is 1D.

Another good example is that any T with $\dim(V)$ distinct eigenvalues must be diagonalizable (each eigenspace has dimension at least 1 by definition).

Our big result for this section is that an operator T is diagonalizable if and only if the minimal polynomial for T is

$$(z - \lambda_1)...(z - \lambda_m)$$

for *distinct* $\lambda_1, ..., \lambda_m \in \mathbb{F}$.

This is another good proof by induction. We only prove the reverse direction. If $m = 1$, $T - \lambda_1 I = 0$ so T is just a multiple of the identity. If $m > 1$ and we have proved the cases $1, ..., m-1$, notice $U = Im(T - \lambda_m I)$ is invariant under T. For any $u \in Im(T - \lambda_m I)$,

$$(T - \lambda_1 I)...(T - \lambda_{m-1} I)u = [(T - \lambda_1 I)...(T - \lambda_{m-1} I)](T - \lambda_m I)v = 0$$

Therefore, $p(z) = (z - \lambda_1)...(z - \lambda_{m-1})$ is a polynomial multiple of the minimal polynomial for $T|_U$. Therefore, by the induction assumption $T|_U$ is diagonalizable and U has a basis of eigenvectors for T. We wish to show that T also looks diagonal off of U. For this, notice that

$$Im(T - \lambda_m I) \cap \ker(T - \lambda_m I) = \{0\}$$

since any vector w in both must have $0 = (T - \lambda_1 I)...(T - \lambda_{m-1} I)w = (\lambda_m - \lambda_1)...(\lambda_m - \lambda_{m-1})w$.

We then have a direct sum $V = Im(T - \lambda_m I) \oplus \ker(T - \lambda_m I)$. The null space is just the eigenspace, and so T clearly has an eigenbasis for that subspace as well.

1. Suppose $T : V \to V$ is diagonalizable and U is invariant under T. Show $T|_U$ is diagonalizable.
2. Let V be a finite-dimensional complex vector space. Show that if $T^n = I$ for any $n \geq 1$, then T is diagonalizable (*Hint:* Roots of Unity).
3. Suppose T is diagonalizable with diagonal matrix A. Show that $\dim(E(\lambda_i, T))$ is the number of times λ_i appears on the diagonal of A
4. Suppose V is finite-dimensional. Show $L(V)$ has a basis consisting of diagonalizable operators *Hint:* Construct a "standard basis" for the matrices, then change the diagonal values when necessary to make them diagonalizable.
5. Show
$$\dim(E(\lambda_1, T)) + ... + \dim(E(\lambda_n, T)) \leq \dim(Im(T))$$
6. Is the matrix
$$\begin{matrix} 11 & 2 \\ 10 & 0 \\ 12 & 1 \end{matrix}$$
diagonalizable?
7. Find T so T^2 is diagonalizable when T is not (*Hint:* Pick the right field).
8. Let A be diagonal with distinct entries on the diagonal and B be any matrix. Show $AB = BA$ if and only if B is diagonal (comes up in the next subsection).

Remark: You should be able to determine when an arbitrary matrix is diagonalizable, even when the matrix is over a weird field like $\mathbb{F}_2$ (this is the same as showing the operator for the matrix is diagonalizable), both using the eigenvalues and eigenvectors and by justifying it with the minimal polynomial. You should be able to compute the diagonal matrix from the eigenvalues, and determine the basis with respect to which it is diagonal (the eigenbasis). One good example to keep in mind is

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

This matrix is small and **not** diagonalizable. Its minimal polynomial is $x^2 - 2x + 1$, which has a repeated root (hence the failure to diagonalize).

13.3 Commuting Operators

Two operators $T, S \in L(V)$ are said to commute if $ST = TS$. Recall that matrices often do not commute, so it should seem interesting when they do. If a pair of matrices are both diagonal, of course they commute, so we already know that when two operators have the same sets fo eigenspaces (and hence the same eigenbasis), they can be diagonalized at the same time and commute. The interesting fact is that two diagonalizable operators commute *if and only if* they are simultaneously diagonalizable.

To prove new direction, notice that if S and T are commuting operators, each eigenspace of S, $E(\lambda, S)$, is invariant under T: $STv = TSv = T(\lambda v) = \lambda Tv$ gives $Tv \in E(\lambda, S)$. Therefore, since S is diagonalizable,

$$V = E(\lambda_1, S) \oplus ... \oplus E(\lambda_n, S)$$

and, using the noted fact, $T|_{E(\lambda_i,S)} \in L(E(\lambda_i, S))$. Because T is diagonalizable, this restriction of T is as well. Therefore, we pick a basis of each $E(\lambda_i, S)$ such that $T|_{E(\lambda_i,S)}$ is diagonal under it. Putting these bases together into a basis for V, T is diagonal. This is also an eigenbasis for S, and so S is diagonal under the basis as well.

If the operators are not diagonalizable, we can't necessarily carry out this argument, but we can see that two commuting operators on a finite-dimensional complex vector space share an eigenvector, and may also simultaneously be written as upper triangular matrices.

This is to say that we still have a strong relationship between the eigenvalues of the two operators.

1. Let S and T be commuting operators on a finite-dimensional complex vector space V. Show that the eigenvalues of $S+T$ are all the sum of an eigenvalue from S and another from T. *Hint:* What are the eigenvalues of an upper triangular matrix?

2. Show that this is not true on a finite-dimensional real vector space.

3. Let V be a finite-dimensional complex vector space. Let $E \subset L(V)$ be a set of diagonalizable operators for which every pair commutes. Show there is a basis of V with respect to which every element of E has a diagonal matrix. *Hint:* Induct on the dimension of V.

14 More General Eigenvectors

[Notice that this skips to Chapter 8 in Axler]

We've already seen some examples of matrices without a full eigenbasis. Consider the case

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

We notice something important: while $A - I$ has a 1-dimension null-space, $(A-I)^2 = 0$, so we may pick some v linearly independent to the eigenvector e_1 and still get some of the "eigenvector properties" from it.

This is actually true for arbitrary operators $T \in L(V)$. We can always fill out some basis for V using this method. To be precise, we define the *generalized eigenspace* for $\lambda \in \mathbb{F}$ to be

$$G(\lambda; T) = \bigcup_{k=0}^{\infty} \ker((T - \lambda I)^k)$$

and the nonzero vectors in this space are called the *generalized eigenvectors*. Notice that $G(\lambda, T)$ is nonzero if and only if λ is an eigenvector (if λ is not an eigenvector, $T(-\lambda I)$ is injective and so are its iterates).

An operator $S \in L(V)$ so $S^n = 0$ is called nilpotent. The reason we can use the above generalized eigenspaces nicely is because vector spaces are a form of "Noetherian Module" (the terminology isn't important here unless you take a lot more math, but these are objects named after the famous algebraist Emmy Noether). Nilpotent operators are quite special, as their minimal polynomial must be of the form z^m for some m and their upper-triangular matrix has zeroes on the diagonal. In particular, there is some basis for V so S has the matrix

$$\begin{bmatrix} 0 & & * \\ & \ddots & \\ 0 & & 0 \end{bmatrix}$$

(why should this be true, given the minimal polynomial?)

Even when an operator is not nilpotent, we can look at the null spaces of iterates of T:

$$\{0\} \subseteq \ker(T) \subseteq \ker(T^2) \subseteq \ker(T^3) \subseteq ...$$

and gain some information. Notice that if $\ker(T^m) = \ker(T^{m+1})$ for any m, then we have $\ker(T^m) = \ker(T^{m+k})$ for any positive k. Now, since V is a finite-dimensional vector space (let $\dim(V) = n$), we have notice that if $\dim(\ker(T^k)) = \dim(\ker(T^{k+1}))$ for $k \leq n$, the chain has already stopped by n. If this doesn't hold, we have

$$0 < \dim(\ker(T)) < \dim(\ker(T^2)) < ... < \dim(\ker(T^n)$$

This implies $\dim(\ker(T^n)) \geq n$, so it must be all of V. In either case, the chain has stopped by n and

$$G(\lambda, T) = ((T - \lambda I)^n)$$

is always enough to compute the generalized eigenspace. Furthermore, since $\ker(T^n) = \ker(T^{n+1})$, we must have that $Im(T^n) \cap \ker(T^n) = \{0\}$, so that applying the FTLM shows

$$V = \ker(T^n) \oplus Im(T^n)$$

[be careful here: this is not true if we remove n].

If we replace T by $(T - \lambda I)$, this looks a lot like a fact we used to prove eigenvalue decompositions. In particular, let V be a finite-dimensional vector space and $T \in L(V)$ have distinct eigenvalues $\lambda_1, ..., \lambda_m$. Then, we claim that

$$V = \bigoplus_{i=1}^{m} G(\lambda_i, T)$$

The proof is by induction on n, the dimension of V. If $n = 1$, this is immediate. Assume we know the fact for all dimensions less than n. We may decompose

$$V = \ker((T - \lambda_1 I)^n) \oplus Im((T - \lambda_1 I)^n) = G(\lambda_1, T) \oplus U$$

Since there is an eigenvector in $G(\lambda_1, T)$, we know $\dim(U) < \dim(V)$. If $U = \{0\}$, we are already done. If not, notice that $U = Im((T - \lambda I)^n)$ is invariant under T since if $u \in U$, $Tu = T(T - \lambda_1 I)^n w = (T - \lambda_1 I)^n (Tw) \in U$.

Therefore, $T|_U \in L(U)$, and applying the induction assumption shows

$$U = G(\lambda_1, T|_U) \oplus ... \oplus G(\lambda_m, T|_U)$$

and so

$$V = \bigoplus_{i=1}^{m} G(\lambda_i, T)$$

One big fact we can glean from this is that we always have a basis of generalized eigenvectors.

For all of these problems, assume V is a finite-dimensional complex vector space.

1. Show that if $v_1 \in G(\lambda_1, T)$ and $v_2 \in G(\lambda_2, T)$ for $\lambda_1 \neq \lambda_2$, then v_1 and v_2 are linearly independent.
2. Let $T \in L(V)$ have $\dim(\ker(T^3)) = 3$ and $\dim(\ker(T^5)) = 4$. Show $\dim(\ker(T^{5+k})) = 4$ for all k.
3. Suppose $T \in L(V)$ is nonzero but nilpotent. Show that T is not diagonalizable.
4. Suppose $\dim(V) = n$ and $\ker(T^{n-1}) \neq \ker(T^n)$. Show T is nilpotent.
5. Find the generalized eigenspaces of

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

14.1 Multiplicity and Characteristic Polynomials

Define the *algebraic multiplicity* of an eigenvalue λ to be

$$\dim(G(\lambda, T))$$

The above decomposition tells us that the sum of these algebraic multiplicities is $\dim(V)$ whenever we are on a complex vector space. If we define the *geometric multiplicity* of an eigenvalue to be

$$\dim(E(\lambda, T))$$

we know that these sum to $\dim(V)$ if and only if T is diagonalizable. Since $E(\lambda, T) \subset G(\lambda, T)$, we are diagonalizable if and only if these multiplicities coincide for each eigenvalue. While this is nice, there is something better we can get from the multiplicities. Let V be a complex vector space, let T have eigenvalues $\lambda_1, ..., \lambda_m$, and let $d_k = \dim(G(\lambda_k, T))$. We define the *characteristic polynomial* (or Cayley polynomial) of T to be

$$p_T(z) = (z - \lambda_1)^{d_1} ... (z - \lambda_m)^{d_m}$$

The degree of the characteristic polynomial is then $\dim(V)$. Furthermore, using the decomposition $V = \bigoplus_{i=1}^{m} G(\lambda_i, T)$, we may find a basis for V such that

$$T = T_1 \oplus ... \oplus T_m$$

and each T_i has a representation as an upper-triangular matrix (with the eigenvalues on the diagonal). Notice that $(T_i - \lambda_i I)^{d_i}|_{G(\lambda_i, T)} = 0$. Therefore,

$$p_T(T)|_{G(\lambda_i, T)} = 0$$

for each i. Because of the decomposition of V, we have that $p_T(T) = 0$, which is known as the Cayley-Hamilton Theorem.

In particular, $p_T(z) = \det(T - zI)$, the polynomial you used in previous linear algebra classes.

1. Let $T \in L(V)$ have *some* upper triangular matrix representation A. Prove that the number of times a value appears on the diagonal of A is its multiplicity as an eigenvalue.

Extends a previous problem Prove that, for invertible $T \in L(V)$,

$$G(\lambda, T) = G(\lambda^{-1}, T^{-1})$$

for every $\lambda \in \mathbb{F}$, $\lambda \neq 0$.

2. Let $T \in L(V)$ have eigenvalues $\lambda_1 \neq \lambda_2$ where both are nonzero. Let $n = \dim(V)$. Show

$$V = \ker(T^{n-2}) \oplus Im(T^{n-2})$$

Hint: You want to show that the chain of kernels of T^k stops by $n - 2$, which is to say that what eigenvalue has multiplicity at most $n - 2$?

3. Given an example of an operator on $\mathbb{C}^n$ whose minimal polynomial is $(z - 1)(z - 2)^2$.

15 Operator Square Roots

A square root of an operator $T \in L(V)$ is another operator S so $S^2 = T$. It is easy to show that not every operator has a square root, even on complex space where we can take roots of negative numbers.

In general, when T is an nilpotent operator, $I + T$ does have a square root. This comes from viewing T under the power series

$$\sqrt{1 + x} = \sum_{n=0}^{\infty} \frac{(-1)^n (2n)!}{(1 - 2n)(n!)^2 4^n} x^n$$

these complicated coefficients don't really matter. What does matter is that this converges for $x \leq 1$.

In a proof for a higher-level math class, we may show directly that (if V has a norm) if we substitute T for x, the sum on the right is an operator on $L(V)$ (though this relies on operator norms and convergence). Let $T^m = 0$. Since we are not assuming we have a norm, instead we notice that if we have

$$S_m = I + a_1 T + ... + a_{m-1} T^{m-1}$$

then

$$S_m^2 = I + 2a_1 T + (2a_2 + a_1^2) T^2 + ...$$

If we pick $a_1 = 1/2$, we see $I + T$ plus some other terms. We then solve for a_2, a_3, and so forth so the higher terms disappear. Eventually, we reach only terms with T^k for $k \geq m$, so those are 0 since T is nilpotent.

On a complex vector space, we can actually say that every operator has a square root since, for eigenvalue λ_k, we can write

$$T|_{G(\lambda_k, T)} = \lambda_k (I + \frac{T - \lambda_k I}{\lambda_k})$$

and use that $T - \lambda_k I$ is nilpotent on the generalized eigenspace to find a square root there.

16 Jordan Normal Form

Remark: Look at the algebra/module type case

17 Trace

18 Inner Products

19 Orthogonality

20 Self-Adjoint and Normal Operators

21 The Spectral Theorem

22 Isometries, Positive Operators

23 Polar Decomposition and Singular Value Decomposition

24 Complexification

25 Normal Operators on a Real Space

26 Multilinear Algebra and the Determinant

27 Solutions to Exercises [New, Not yet complete]

27.1 Quotients

1.) Consider the set of vectors $\{(v_i, 0), (0, w_j)\}_{i=1,j=1}^{i=m,j=n}$. We notice that for an (v, w) in the product, $v = \sum a_i v_i$ and $w = \sum b_j w_j$ means

$$(v, w) = \sum a_i (v_i, 0) + \sum b_j (0, w_j)$$

so that this set spans the product. Further, if

$$\sum \alpha_i(v_i, 0) + \sum \beta_j(0, w_j) = (0, 0)$$

then

$$\sum \alpha_i v_i = 0$$
$$\sum \beta_j w_j = 0$$

so that the coefficients are all 0 using the linear independence of these vectors. This checks that the collection $\{(v_i, 0), (0, w_j)\}_{i=1,j=1}^{i=m,j=n}$ is also linearly independent, and hence a basis.

2.) The matrix for $T \times S$ is the block matrix

$$\begin{bmatrix} M(T) & 0 \\ 0 & M(S) \end{bmatrix}$$

(you should justify this).

3.) Consider $T \in L(V, W_1 \times W_2)$. We can construct $T_1 \in L(V, W_1)$ and $T_2 \in L(V, W_2)$ by taking the projections $\pi_1(x, y) = x$, and $\pi_2(x, y) = y$, then setting

$$T_1 = \pi_1 \circ T$$
$$T_2 = \pi_2 \circ T$$

Notice that $Tx = (0, 0)$ if and only if $T_1(x) = 0$ and $T_2(x) = 0$. Therefore, the map $T \mapsto (T_1, T_2)$ is injective. Given $S_1 \in L(V, W_1)$ and $S_2 \in L(V, W_2)$, one may also note that this association takes $Sv := (S_1 v, S_2 v) \mapsto (S_1, S_2)$, and so it is surjective. Lastly, the map is linear and thus an isomorphism.

4.) Using problem 3,

$$V^m \simeq (L(\mathbb{F}, V))^m \simeq L(\mathbb{F}^m, V)$$

5.) In the text.

6.) Take the lifts of the quotient basis $\{v_1, ..., v_n\}$ in V. We notice that for an arbitrary $v \in V$, $v + U = \sum a_i(v_i + U) = (\sum a_i v_i) + U$. This implies that $v - \sum a_i v_i \in U$. Thus,

$$v - \sum a_i v_i = \sum b_j u_j$$

or

$$v = \sum a_i v_i + \sum b_j u_j$$

and so these vectors span V. Furthermore, if

$$\sum \alpha_i v_i + \sum \beta_j u_j = 0$$

we must have by applying the quotient map that $\sum(\alpha_i v_i + U) = 0 + U$. This implies that $\alpha_i = 0$ for all i by the definition of the v_i. Hence, $\sum \beta_j u_j = 0$, which implies the β_j are 0 since the u_j are linearly independent. We thus have that the given collection is also linearly independent.

7.) Since V/U is finite-dimensional, it has a basis $\{v_i + U\}_{i=1}^m$. Consider the lifts $v_i \in V$. For arbitrary $w \in V/U$, we have $w = \sum w_i v_i + U$. We construct a map $\phi : V/U \times U \to V$ given by

$$\phi(w + U, u) = u + \sum w_i v_i$$

First, this map is an injection since $\phi(w+U, u) = u+\sum w_i v_i = 0$ implies $\sum w_i v_i \in U$, or that $\sum w_i v_i + U = 0 + U$. Hence, our assumption that the $v_i + U$ were a basis implies $w_i = 0$ for all i. Therefore, $u + 0 = 0$, and $u = 0$ as well. Second, the map is a surjection because any $v \in V$ has $v + U = \sum a_i v_i + U$ and $v - \sum a_i v_i \in U$, so that

$$\phi(\sum a_i v_i + U, v - \sum a_i v_i) = v$$

We then have the claimed isomorphism.

8.) Using the same notation as above, $W = \text{span}\{v_i\}$ works.

27.2 Duality

27.3 Upper Triangular

27.4 Diagonal Matrices

27.5 Commuting operators

1.) Pick a basis under which both S and T have upper triangular matrices. Then, $M(S+T) = M(S)+M(T)$, and the diagonal values are the eigenvalues. Since the diagonal values of $M(S+T)$ are then the sum of an eigenvalue of S and another of T, we have the claim.

2.) The matrices

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

clearly commute. While A and $A+B$ have real eigenvalues, B does not.

3.) In the case $\dim(V) = 1$, all operators are diagonal. Assume we have shown that the claim holds for $\dim(V) < n$. Next, if E is empty, we are done. Furthermore, if every $S \in E$ has only one eigenvalue, then they are all scalar multiples of the identity and we are done. Therefore, we may pick $S \in E$ with $m \geq 2$ distinct eigenvalues.

Since S is diagonalizable,

$$V = E(\lambda_1, S) \oplus ... \oplus E(\lambda_k, S)$$

For any $T, T' \in E$, $T|_{E(\lambda_i,S)}, T'|_{E(\lambda_i,S)} \in L(E(\lambda_i, S))$, and these two operators commute since they commute on V.

Define E' to be the set of $T|_{E(\lambda_i,S)}$ for $T \in E$. Since $\dim(E(\lambda_i, S)) < n$, we have by the induction hypothesis that there is a basis $\{v_1^i, ..., v_{n_i}^i\}$ for $E(\lambda_i, S)$ under which every operator in E' has a diagonal matrix. Combining these bases,

$$\{v_1^1, ..., v_{n_1}^1, v_1^2 ... v_{n_2}^2 ... v_{n_m}^m\}$$

is a basis under which every operator in E has a diagonal matrix.

27.6 Generalized Eigenvectors

1.) Suppose v_1 and v_2 are given, such that $v_i \in \ker(T - \lambda_i I)^{k_i}$ (assume these are the smallest k_i where this holds). By contradiction, assume $v_1 = \alpha v_2$ for some α. Then,

$$(T - \lambda_1 I)^{k_1} v_1 = 0 = (T - \lambda_2 I)^{k_2} v_2$$

However, then

$$0 = ((T - \lambda_1 I) + (\lambda_1 I - \lambda_2 I))^n v$$
$$= \sum_{k=0}^{n} \binom{n}{k} (\lambda_1 - \lambda_2)^k (T - \lambda_1 I)^{n-k} v$$

If we apply $(T - \lambda_1 I)^{k_1 - 1}$ to both sides, we see that $= 0(\lambda_1 - \lambda_2)^n (T - \lambda_1 I)^{k_1 - 1} v$, a contradiction as desired.

2.) Notice that $\ker(T^3) \subset \ker(T^4) \subset \ker(T^5)$. We consider two possibilities. If $\ker(T^4)$ is 3-dimensional, the inclusion implies $\ker(T^4) = \ker(T^3)$. However, this implies $\ker(T^5) = \ker(T^4)$, so this case is impossible using the assumed dimension. In the other case, $\ker(T^4)$ is 4-dimensional and $\ker(T^4) = \ker(T^5)$ similarly. Again, this implies $\ker(T^{5+k}) = \ker(T^5)$ for all $k \geq 0$.

3.) The minimal polynomial for T must be of the form z^k since T is nilpotent. However, T is nonzero so $k > 1$. Since the minimal polynomial doesn't have linear factors, T is not diagonalizable.

4.) We know that $\ker(T^{n+k}) = \ker(T^n)$ for all $k \geq 0$, or that the chain

$$0 \subset \ker(T) \subset \ker(T^2) \subset ...$$

stops by n. By the assumption of the problem, it cannot stop before n, so $\dim(\ker(T^k)) < \dim(\ker(T^{k+1}))$ for $k < n$. This implies

$$0 < \dim(\ker(T)) < ... < \dim(\ker(T^n)) \leq n$$

where the last inequality holds because this is a subspace of V. Then, $\dim(\ker(T^m)) = m$ for each $m \leq n$ and the last kernel must be all of V.

5.) We notice that the eigenvalues of this matrix are 1 and 2. First,

$$(A - I)^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

has the same null space as $A - I$, so the generalized eigenspace here is just $\text{span}\{e_1\}$. However,

$$(A - 2I)^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

has 2D null space $\text{span}\{e_2, e_3\}$. Further powers of this matrix are the same, so we are done.